

Report Data Integration & Application

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01-07-2026

▼ Part 1 - Data Profiling and Data Quality Analysis

▼ A - Dataset Structure

Dataset	Column Name	Data Type	Semantic Meaning
Caractéristiques (Characteristics)	Num_Acc	int64	Unique accident identifier
	jour	int64	Day of the accident
	mois	int64	Month of the accident
	an	int64	Year of the accident
	hrmn	object	Time of the accident (hours and minutes)
	lum	int64	Lighting conditions (e.g., daylight, night with/without public lighting)
	dep	object	Department code (INSEE code)
	com	object	Municipality (Commune) INSEE code
	agg	int64	Localization: In or out of built-up areas
	int	int64	Intersection type (e.g., T-junction, roundabout, crossroads)
	atm	int64	Atmospheric conditions (e.g., rain, snow, fog, clear)
	col	int64	Collision type (e.g., frontal, rear-end, multiple collisions)
	adr	object	Postal address (for accidents in built-up areas)
	lat	object	Latitude coordinate
	long	object	Longitude coordinate
Lieux (Locations)	Num_Acc	int64	Unique accident identifier
	catr	int64	Category of road (e.g. highway, national road, street)

Dataset	Column Name	Data Type	Semantic Meaning
	voie	object	Road number
	v1	int64	Numerical index of the road number
	v2	object	Alphanumeric index letter of the road
	circ	int64	Traffic regime (one-way two-way, separated lanes)
	nbv	object	Total number of traffic lanes
	vosp	int64	Presence of a reserved lane (bike lane, bus lane)
	prof	int64	Longitudinal profile describing the road slope/gradient
	pr	object	Landmark/Milestone number
	pr1	object	Distance in meters to the milestone marker
	plan	int64	Plan view layout/alignment of the road
	lartpc	object	Width of the central reservation / median strip
	larrout	object	Width of the roadway allocated to vehicle
	surf	int64	Surface condition of the road (wet, icy, dry, muddy)
	infra	int64	Infrastructure features
	situ	int64	Location situation of the accident (roadway, sidewalk)
	vma	int64	Maximum speed limit at the time and place of the accident
Véhicules (Vehicules)	Num_Acc	int64	Unique accident identifier
	id_vehicules	object	Unique numerical identifier for the vehicle
	num_veh	object	Alphanumeric vehicle identifier
	senc	int64	Direction of travel (increasing/decreasing milestone, postal number)
	catv	int64	Category of the vehicle
	obs	int64	Fixed obstacle struck
	obsm	int64	Mobile obstacle struck

Dataset	Column Name	Data Type	Semantic Meaning
	choc	int64	Initial point of impact
	manv	int64	Main maneuver before the accident (overtaking, turning, changing lanes)
	motor	int64	Type of vehicle motorization
	occutc	float64	Number of passengers in the public transportation vehicle
Usagers (Users)	Num_Acc	int64	Unique accident identifier
	id_usager	object	Unique numerical identifier of user
	id_vehicule	object	Unique numerical identifier of the vehicle the user occupied
	num_veh	object	Alphanumeric identifier of the vehicle the user occupied
	place	int64	Seating position occupied in/on the vehicle by the user
	catu	int64	User category (1: Driver, 2: Passenger, 3: Pedestrian)
	grav	int64	Severity of the injury (1: Unharmed, 2: Killed, 3: Hospitalized injured, 4: Lightly injured)
	sexe	int64	User gender
	an_nais	float64	Birth year of the user
	trajet	int64	Purpose of travel at the time of the accident
	secu1	int64	Use and presence of the primary safety equipment
	secu2	int64	Use and presence of the primary safety equipment
	secu3	int64	Use and presence of the primary safety equipment
	locp	int64	Location of the pedestrian
	actp	object	Action of the pedestrian
	etatp	int64	State of the pedestrian (Alone, accompanied, in group)

Remarks

- The attribute `an_nais` should be an integer and not a float number, according to its description.
- The attribute `occutc` should be an integer and not a float number, according to its description.
- The attributes `lat` and `long` would have made more sense in float, but we need to check the actual content of its values to confirm.

▼ B - Missing values and completeness

Missing values can be represented by the values null, NaN (possibly as the value or even the string) and -1 (which corresponds to 'No information').

Dataset	Column	Missing %	Comment
Caractéristiques (Characteristics) 0.28%	Num_Acc	0	
	jour	0	
	mois	0	
	an	0	
	hrmn	0	
	lum	0	
	dep	0	
	com	0	
	agg	0	
	int	0	
	atm	0	
	col	0.01	Negligeable loss, the rows should be dropped since it's not a value that can be easily replaced.
	adr	4.25	Negligeable loss, the rows should be dropped since it's not a value that can be easily replaced.
	lat	0	
	long	0	
Lieux (Locations) 13.99%	Num_Acc	0	
	catr	0	

Dataset	Column	Missing %	Comment
	voie	18.98	The amount of missing value is not that high but it can be an important information. Should just standardize the value of missing values.
	v1	23.16	The amount of missing value is not that high but it can be an important information. Should just standardize the value of missing values.
	v2	91.58	Too much loss and the feature is not critical - it should be dropped entirely.
	circ	6.20	Negligeable loss, the rows should be dropped since it can be interesting to analyze.
	nbv	0	
	vosp	5.45	Negligeable loss, the rows should be dropped since it can be interesting to analyze.
	prof	0.07	Negligeable loss, the rows should be dropped since it can be interesting to analyze.
	pr	0	
	pr1	0	
	plan	0.06	Negligeable loss, the rows should be dropped.
	lartpc	99.95	Too much loss and the feature is not critical - the column should be dropped entirely.
	larrout	0	
	surf	0.05	Negligeable loss, the rows should be dropped.
	infra	1.16	Negligeable loss, the rows can be kept as it is since it's not necessarily a critical information.
	situ	0.05	Negligeable loss, the rows should be dropped.
	vma	5.17	Negligeable loss, the rows should be dropped because it's important to see if there are certain

Dataset	Column	Missing %	Comment
			limitations that are more prone to accidents than others.
Véhicules (Vehicules) 9.04%	Num_Acc	0	
	id_vehicules	0	
	num_veh	0	
	senc	0.07	Negligeable loss, the rows can kept as it is since it's not a critical information.
	catv	0	
	obs	0.03	Negligeable loss, the rows should be dropped.
	obsm	0.03	Negligeable loss, the rows should be dropped.
	choc	0.05	Negligeable loss, the rows should be dropped to understand the hit zone frequency.
	manv	0.03	Negligeable loss, the rows should be dropped because it would be too important to see if movements can impact the outcome.
	motor	0.21	Negligeable loss, the rows can be dropped.
	occutc	98.98	Too much loss and the feature is not critical since it's difficult to get the number of
Usagers (Users) 17.64%	Num_Acc	0	
	id_usager	0	
	id_vehicule	0	
	num_veh	0	
	place	0	
	catu	0	
	grav	0	
	sexe	1.91	Negligeable loss, the rows can kept as it is since it's not a critical information.
	an_nais	2.06	Negligeable loss, the rows should be dropped to be able to analyze

Dataset	Column	Missing %	Comment
			an age demographic tendency.
	trajet	2.10	Negligeable loss, the rows can kept as it is since it's not a critical information.
	secu1	1.68	Negligeable loss, the rows can kept as it is since it's not a critical information.
	secu2	42.99	A lot of loss on an information that is not that critical - the column should be dropped.
	secu3	90.37	Too much loss on an information that is not that critical - the column should be dropped.
	locp	49.34	There is a lot of loss but by default we could transform the value to '0 - No pedestrian'
	actp	0	
	etatp	91.81	Too much loss on an information that is not that critical - the column should be dropped.

▼ C - Consistency and validity checks

→ Value ranges & categorical anomalies

Dataset	Column w. anomalies	Anomaly Description
Lieux (Location)	vma	The attribute (considering only valid ones) ranges from 0 to 900 which is impossible. The valid range is between 30 and 130, but 6.22% of rows is outside of it. These corresponding rows should be dropped since this attribute is rather critical.

→ Duplicates

Dataset	Number of duplicates	Action
Lieux (Location)	2	Drop the duplicated line, there is only one
Caractéristiques (characteristics)	0	
Véhicules (vehicles)	0	
Usagers (users)	0	

▼ D - Data Quality Summary

While duplicated rows are negligible in this dataset, several critical data issues directly impact our analytical workflow:

- **Inconsistent Data Types**: Multiple attributes are assigned incorrect data types that contradict the official dataset description (e.g., categorical codes or coordinates stored as generic text object types instead of integers or floats).
⇒ This prevents us from executing direct mathematical operations, filtering ranges efficiently, or feeding features directly into our machine learning models without extensive preprocessing.
- **Prevalent Missing and Null Values**: Several features consist almost entirely of missing data, empty strings, zeros (or -1s in our case).
⇒ This can reduce the statistical power of these variables. We need to decide whether to drop these columns entirely or implement strategies to avoid model predictions with low performance (dropping the affected rows, replacing by the mean or median etc.)
- **Out-of-Range and Invalid Values**: Multiple records contain values that fall outside the logical boundaries or official ranges defined in the documentation.
⇒ These anomalies act as noise, distorting basic statistical metrics (like means and variances) and introducing bias into optimization landscapes or classification boundaries.

▼ Part 2 - Transformations, Modeling and Medallion Architecture

▼ A - Required Transformations (silver layer)

1 - Standardization: Format Normalization

- **Temporal Features** : Combine the separate year, month, and day columns, along with the `hrmn` (hours and minutes string) column, into a single unified `datetime64[ns]` timestamp. This standardizes the temporal axis and enables time-series operations.
- **Geospatial Coordinates** : Transform `lat` (latitude) and `long` (longitude) text fields from generic `object` types into `float64` numerical values, ensuring coordinates adhere to standard decimal degrees.
- **Categorical Codes**: Standardize missing data representation by replacing all disparate placeholder formats with a uniform `NaN` value across all four tables.

- **Data Type Casts:** Convert demographic variables like `an_nais` (birth year) and public transport occupancy `occutc` from `float64` or `object` types to standard 32-bit integers to match their logical definitions.

2 - Cleaning: Invalid and Missing Value Management

- **Handling High-Sparsity Features:** Drop features that contains critically high missingness rates where information cannot be reliably replaced:
 - `v2` (91.58% missing)
 - `lartpc` (99.95% missing)
 - `occutc` (98.98% missing)
 - `secu2` (42.99% missing) and `secu3` (90.37% missing)
 - `etatp` (91.81% missing)
- **Conditional Imputation:** For the pedestrian location feature `locp` (49.34% missing), replace missing values with a new category code `0` to explicitly represent "No pedestrian involved", preventing unnecessary row loss.
- **Incorrect Boundary Filtering (`Lieux`):** Address the maximum authorized speed (`vma`) anomalies where values range up to 900 km/h. Drop the 6.22% of rows containing values outside the legal French road speed limits (outside the [30, 130] km/h bracket) , as this variable is critical to classification accuracy and cannot be arbitrarily imputed without introducing structural bias.
- **Deletion for Low-Sparsity Anomalies:** Apply listwise deletion (dropping rows) for minimal, isolated missing data across critical features like `col` (0.01%) , `circ` (6.20%) , `surf` (0.05%) , and `manv` (0.03%) to ensure downstream multivariate statistical integrity.

3 - Enrichment: Derived Feature Engineering

- **Accident Severity Index (Target Variable Engineering):** Synthesize a global record-level classification target by aggregating individual user outcomes from the `Usagers` table.
- **Time-of-Day Categorization:** Parse the engineered timestamp or the `hrmn` column to create a new categorical feature partitioning the day into distinct traffic regimes (e.g., *Morning Peak*, *Afternoon Slack*, *Evening Peak*, *Night*). This provides a more distinct behavioral indicator than a raw continuous time string.
- **User Age Demographic:** Compute a continuous `age` feature by subtracting the user's birth year (`an_nais`) from the accident year (`an`), providing a direct demographic variable for variance analysis.

4 - Deduplication

- **Targeted Record Elimination (**Lieux**):** Execute a deduplication pass targeting the **Lieux** dataset to remove the 2 exact duplicate rows discovered during profiling.